

Electronics Homework 3 : MOS Amplifier Design

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June 16, 2026

Design Specifications and Goals:

- $V_{DD} = 3V$, $R_{sig} = 3k\Omega$, $R_L = 10k\Omega$.
- Coupling and Bypass Capacitors: $CCI = CCO = CS = 5\mu F$.
- Process: $0.5\mu m$ CMOS technology (SPICE Level-1 Model).
- Targets: Gain $|v_o/v_{sig}| \geq 30$, and power consumption $< 1mW$.

Problem a (*Design of Bias Current and W/L Ratios*)

Design Parameters and Assumptions Before performing the detailed design, we establish the fundamental parameters based on the $0.5\mu m$ CMOS process provided in Table 1. The key constants used for our calculations are as follows:

- **Process Constants:** $\mu_n = 460 \text{ cm}^2/V \cdot s = 0.046 \text{ m}^2/V \cdot s$, $t_{ox} = 9.5 \text{ nm}$.
- **Oxide Capacitance:** $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.854 \times 10^{-12}}{9.5 \times 10^{-9}} \approx 3.63 \times 10^{-3} \text{ F/m}^2$.
- **Transconductance Parameter:** $k'_n = \mu_n C_{ox} \approx 167 \times 10^{-6} \text{ A/V}^2$.

1. Initial Power Consumption Design To ensure the total power consumption $P_{total} < 1mW$ under $V_{DD} = 3V$, the total current is constrained:

$$I_{total} = I_D + I_{bias} + I_{bias.network} < \frac{1mW}{3V} \approx 333\mu A.$$

We select $I_D = 100\mu A$, $I_{bias} = 100\mu A$, and $(R_{G1} + R_{G2}) = 2M\Omega$. Including the biasing network current $I_{bias.network} = V_{DD}/(R_{G1} + R_{G2}) = 3V/2M\Omega = 1.5\mu A$, the total current is:

$$I_{total} = 100\mu A + 100\mu A + 1.5\mu A = 201.5\mu A.$$

The resulting total power consumption is:

$$P = V_{DD} \times I_{total} = 3V \times 201.5\mu A = 0.6045mW \quad (\text{Satisfies } P < 1mW).$$

2. DC Biasing and Resistor Design (R_D, R_S)

Following the requirement $V_{RD} = V_{DS1} = V_{DD}/3 = 1V$:

- **Drain Resistor (R_D):** $R_D = \frac{V_{RD}}{I_D} = \frac{1V}{100\mu A} = 10k\Omega$.
- **Source Resistor (R_S):** To satisfy $V_{DS1} = 1V$ under $V_{DD} = 3V$ and $V_{RD} = 1V$, the source voltage must be $V_S = V_{DD} - V_{RD} - V_{DS1} = 3V - 1V - 1V = 1V$. Thus: $R_S = \frac{V_S}{I_D} = \frac{1V}{100\mu A} = 10k\Omega$.
- **Verification:** $V_{D1} = V_{DD} - I_D R_D = 2V$, and $V_{DS1} = V_{D1} - V_S = 1V$ (Exactly matches the specification).

3. Design of M1 (W/L) for Gain Requirement

The gain is $A_v \approx -g_{m1}(R_D \parallel R_{in,M2})$. Since $R_{in,M2} \gg R_D$, we simplify to $A_v \approx -g_{m1}R_D$. With C_S providing an AC short to ground, source degeneration is bypassed, enabling maximum gain. Thus, $g_{m1} = \frac{30}{10k\Omega} = 3mA/V$. Using the transconductance formula $g_{m1} = \sqrt{2k'_n(W/L)I_D}$:

$$(3 \times 10^{-3})^2 = 2 \times (167 \times 10^{-6}) \times \left(\frac{W}{L}\right) \times 100 \times 10^{-6}$$

$$9 \times 10^{-6} = 33.4 \times 10^{-9} \times \left(\frac{W}{L}\right) \implies \frac{W}{L} \approx 269.$$

With $L_1 = 0.5\mu m$, we obtain $W_1 = 134.5\mu m$.

4. Design of M2 (W/L) Buffer Stage

To ensure low output impedance, we require $1/g_{m2} \ll R_L$ ($10k\Omega$). Setting $1/g_{m2} = 1k\Omega$ gives $g_{m2} = 1mA/V$:

$$(1 \times 10^{-3})^2 = 2 \times (167 \times 10^{-6}) \times \left(\frac{W_2}{L}\right) \times 100 \times 10^{-6}$$

$$1 \times 10^{-6} = 33.4 \times 10^{-9} \times \left(\frac{W_2}{L}\right) \implies \frac{W_2}{L} \approx 30.$$

With $L_2 = 0.5\mu m$, we obtain $W_2 = 15\mu m$.

5. Design of Biasing Resistors (R_{G1}, R_{G2})

First, calculate V_{GS1} from $I_D = \frac{1}{2}k'_n(W/L)(V_{GS} - V_{th})^2$:

$$100\mu = 0.5 \times 167\mu \times 269 \times (V_{GS} - 0.7)^2 \implies V_{GS} \approx 0.766V.$$

Setting $V_G = V_{GS} + V_S = 0.766 + 1 = 1.766V$. Using the divider rule with $R_{G1} + R_{G2} = 2M\Omega$:

$$V_G = 3V \times \frac{R_{G2}}{2M\Omega} = 1.766V \implies R_{G2} \approx 1.177M\Omega, R_{G1} \approx 823k\Omega.$$

Conclusion The design successfully achieves the required gain and power constraints.

The finalized parameters are summarized below:

Component	Value	Component	Value
$M1$	$W = 134.5\mu, L = 0.5\mu$	R_D	$10k\Omega$
$M2$	$W = 15\mu, L = 0.5\mu$	R_S	$10k\Omega$
R_{G1}	$823k\Omega$	R_{G2}	$1.177M\Omega$

Problem b (Frequency Response Analysis)

1. Circuit Schematic The circuit schematic used for the frequency response simulation is shown below:

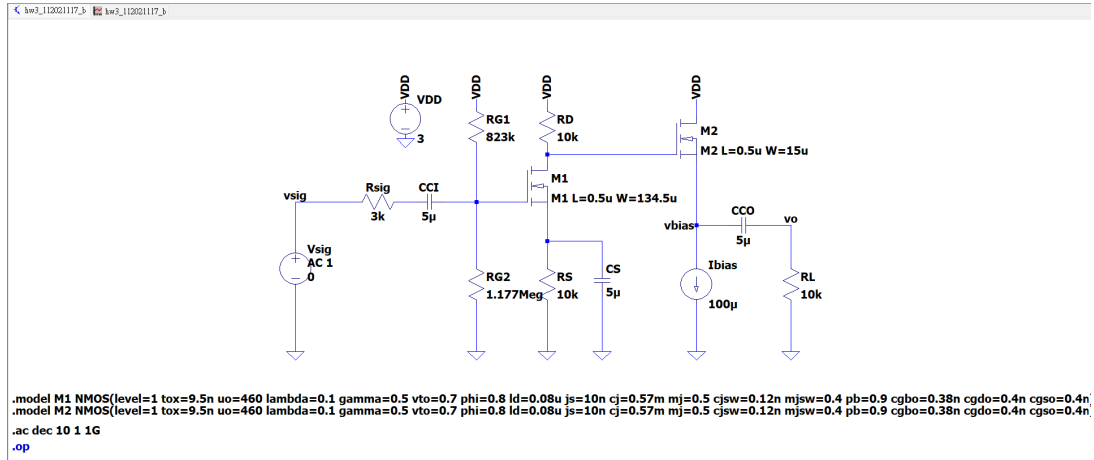


Figure 1: Frequency response analysis circuit schematic.

2. Simulation Results and Gain Verification An AC sweep from 1Hz to 1GHz was performed. As shown in the plot below, the midband gain is approximately 30.17dB, which satisfies the design requirement ($A_v \geq 30$, i.e., 29.54dB).

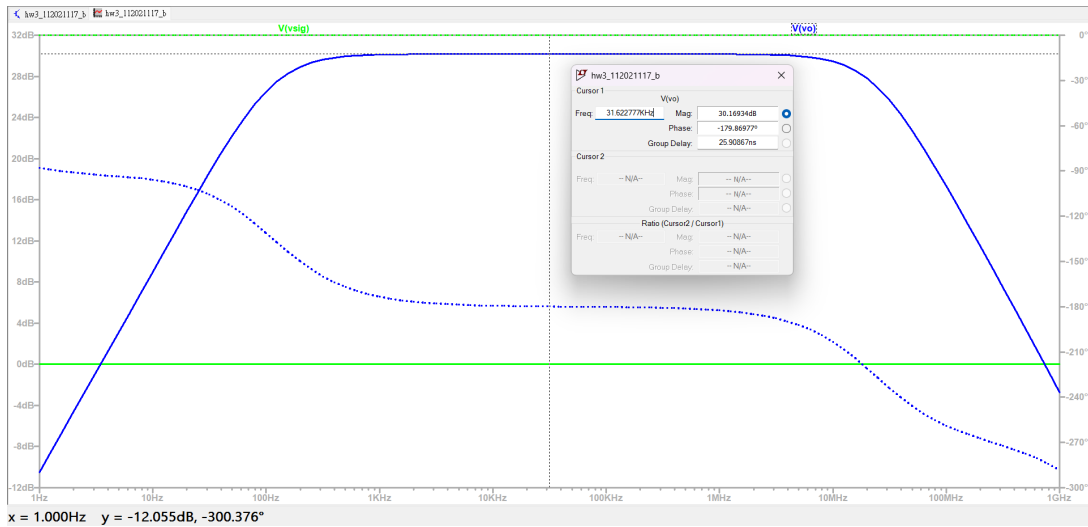


Figure 2: Midband gain verification.

3. Bandwidth Analysis (3-dB Frequencies) The 3-dB cutoff frequencies were identified by marking the frequencies where the gain drops by 3dB from the midband value.

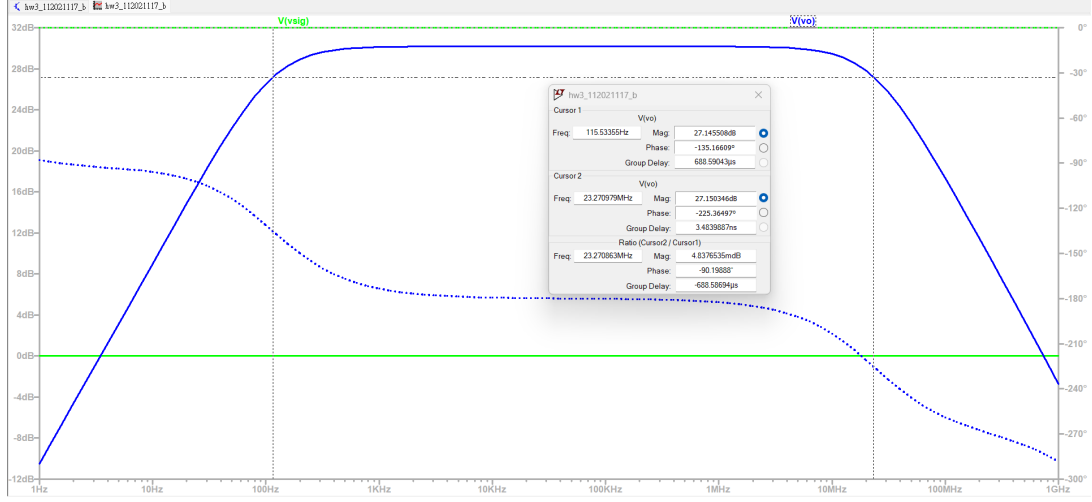


Figure 3: Identification of lower and upper 3-dB frequencies.

Summary of Results

Parameter	Value
Midband Gain	30.17dB
Lower 3-dB Frequency (f_L)	115.5Hz
Upper 3-dB Frequency (f_H)	23.27MHz
Bandwidth ($f_H - f_L$)	$\approx 23.27\text{MHz}$

Analysis The simulation confirms a stable gain across a wide bandwidth:

- **Low-Frequency Cutoff ($f_L \approx 115.5\text{Hz}$):** Established by the coupling/bypass capacitors (C_{CI}, C_S, C_{CO}).
- **High-Frequency Cutoff ($f_H \approx 23.27\text{MHz}$):** Limited by the internal parasitic capacitances of the MOSFETs.

4. Verification of Total Power Consumption The total power consumption is calculated using the total current drawn from the V_{DD} source, obtained from the DC operating point (.op) simulation. This value accounts for the current in all branches, including the common-source stage, the source-follower buffer, and the biasing network. From the simulation output:

$$I_{total} = |I(V_{DD})| = 202.77\mu\text{A}.$$

The actual total power consumption is:

$$P_{actual} = V_{DD} \times I_{total} = 3\text{V} \times 202.77\mu\text{A} \approx \mathbf{0.6083\text{mW}}.$$

This confirms that the total power dissipation is well within the design specification ($P < 1\text{mW}$).

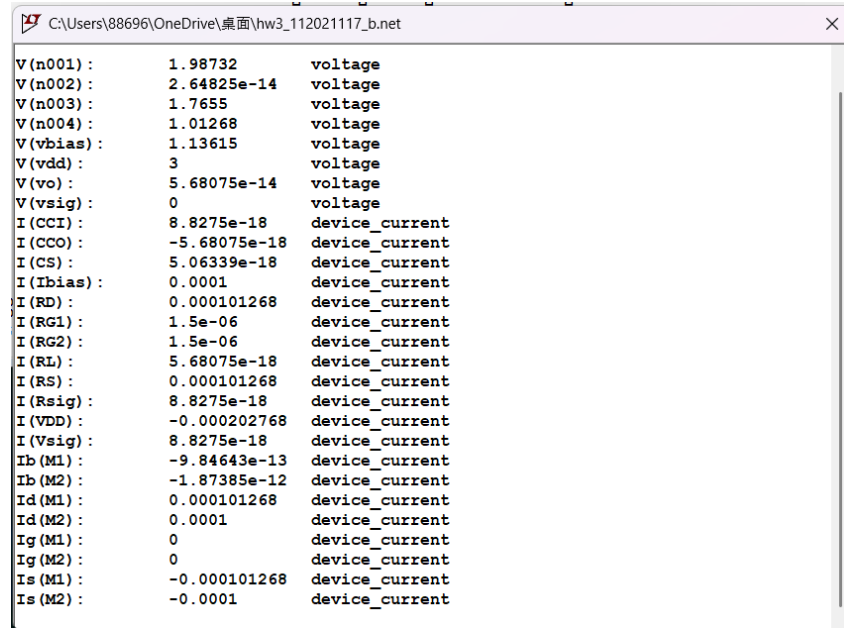


Figure 4: DC Operating Point (.op) analysis result showing the total current $I(V_{DD})$.

Problem c (Dynamic Range Analysis)

1. Circuit Schematic and Operating Point The circuit configuration and DC operating point data are shown below:

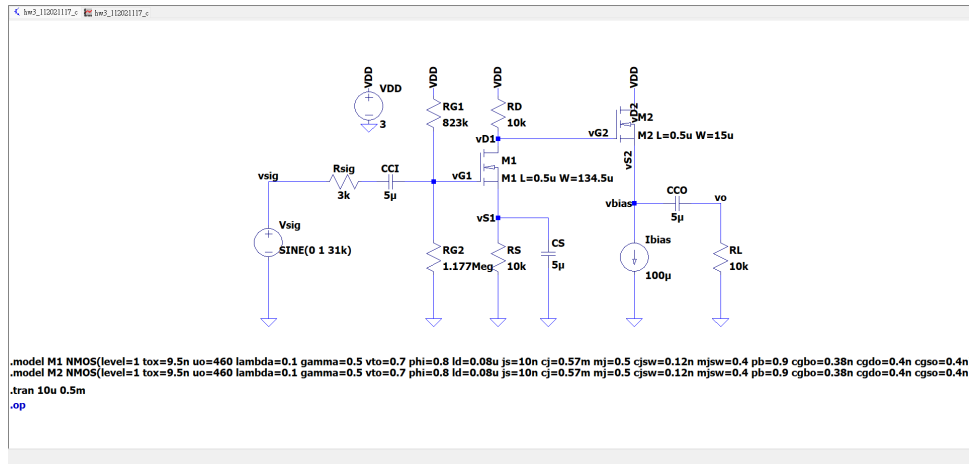


Figure 5: Circuit Schematic

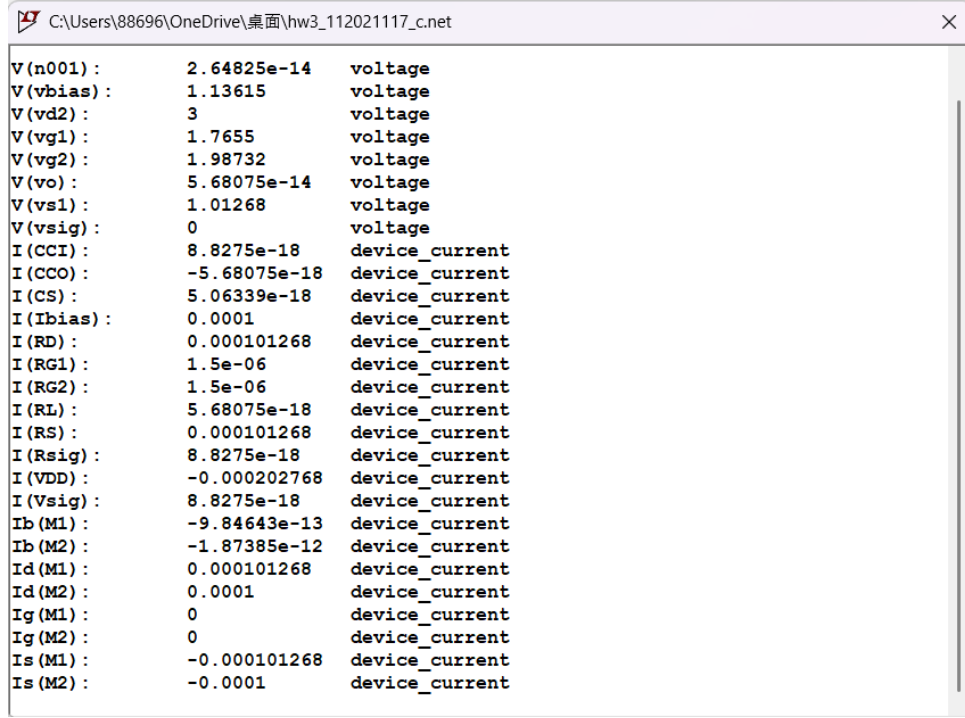


Figure 6: DC Operating Point (.op) Data

2. Hand Calculation Based on the DC bias conditions, the theoretical limits of the linear swing are calculated as follows:

- **Circuit Parameters:**

- $V_{DS1} = V_{DD} - I_D(R_D + R_S) = 3V - 0.1mA(20k\Omega) = 1.0V$
- $V_{GS1} = V_{G1} - V_{S1} = 1.7655V - 1.01268V = 0.7528V$
- $V_{OV1} = V_{GS1} - V_{th} = 0.7528V - 0.7V = 0.0528V$
- $g_{m1} = \frac{2I_D}{V_{OV1}} \approx 3.79mS$
- $R_{out} = R_D \parallel R_L = 5k\Omega$

- **Output Swing Analysis:**

- Saturation Boundary (v_{o+}):

$$\hat{v}_{o+} = V_{DS1} - (V_{GS1} - V_{th}) = 1.0V - 0.0528V = 0.9472V$$

- Cut-off Constraint (v_{o-}):

$$\hat{v}_{o-} = V_{OV1} \cdot g_{m1} \cdot R_{out} = 0.0528V \cdot 3.79mS \cdot 5k\Omega \approx 1.000V$$

- Theoretical Maximum Output Swing:

$$\hat{v}_{o,max} = \min(\hat{v}_{o+}, \hat{v}_{o-}) = \mathbf{0.9472V}$$

3. Simulation Result vs. Hand Calculation The transient simulation result is shown in Figure 3:

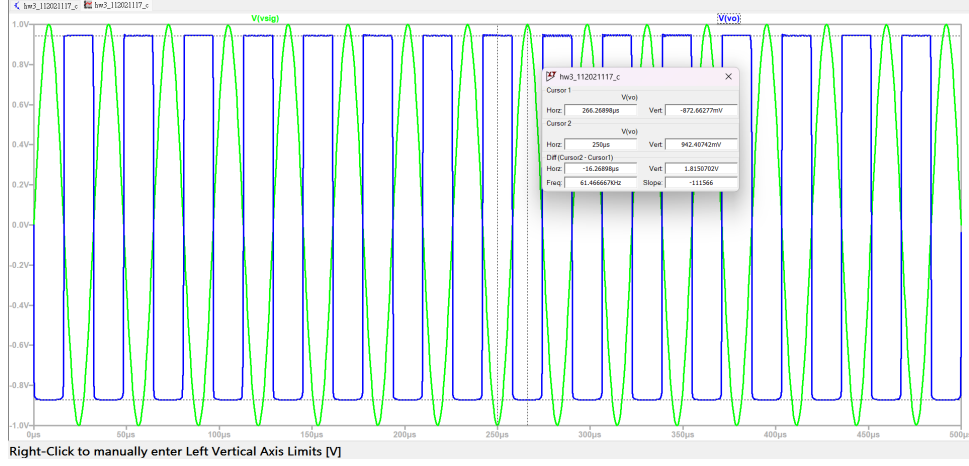


Figure 7: Transient Simulation of Output Voltage (Negative swing limit $\approx -0.871\text{V}$)

- Analysis: The theoretical maximum output swing is 0.947V , while the simulated limit is approximately 0.871V . This deviation of 76mV is primarily attributed to the Body Effect, which increases the threshold voltage (V_{th}) in practice, and the non-ideal voltage level shift and non-linear behavior of the Source Follower ($M2$) under large signal swings.

4. Input Dynamic Range Calculation Using the midband gain $|A_v| \approx 32.25$ and the simulated output limit of 0.871V :

$$\hat{v}_{sig,max} = \frac{\hat{v}_{o,max,sim}}{|A_v|} = \frac{0.871\text{V}}{32.25} \approx 27.01\text{mV}$$

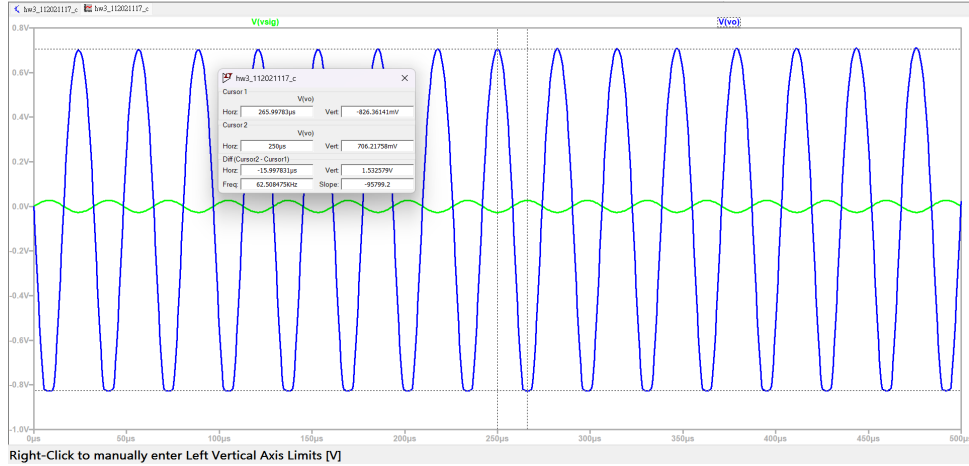


Figure 8: Input signal v_{sig} swing limit simulation

Problem d (*Influence of Transistor M2*)

1. Circuit Schematic and DC Analysis By removing the transistor M2 (Source Follower), the circuit is simplified to a standard Common Source (CS) amplifier, with the load (R_L and C_{CO}) directly coupled to the drain of M1.

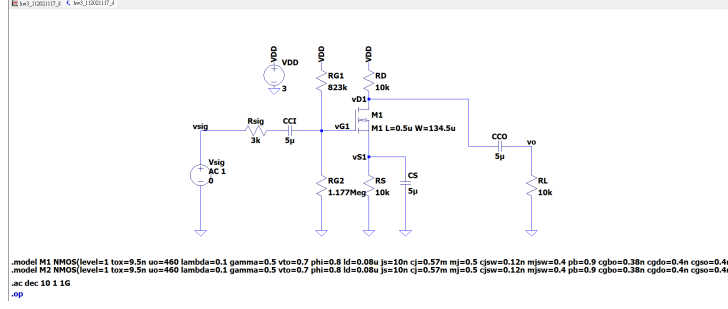


Figure 9: Circuit schematic with M2 removed.

From the .op analysis, the DC operating point of M1 is identified as $I_D \approx 101.27\mu\text{A}$.

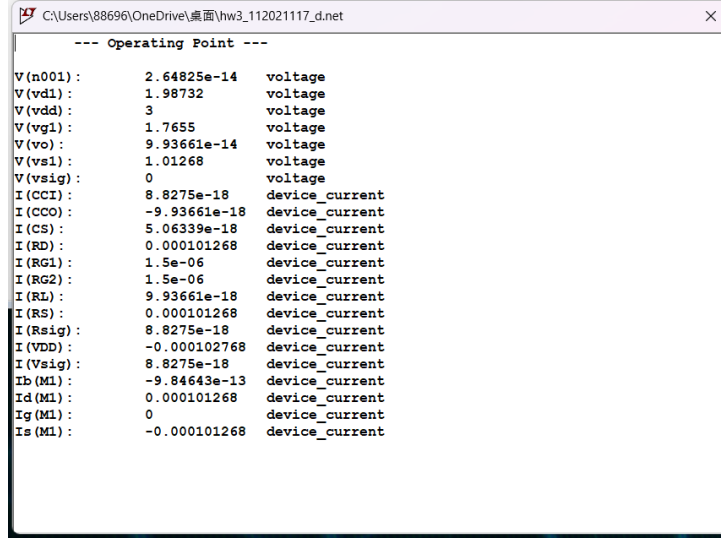


Figure 10: DC Operating point analysis for the M2-removed circuit.

2. Quantitative Midband Gain Calculation

- **Operating Point Parameters:** From the .op analysis, we identify $V_{G1} = 1.7655\text{V}$ and $V_{S1} = 1.01268\text{V}$, yielding $V_{GS1} \approx 0.753\text{V}$. With $V_{th} = 0.7\text{V}$, the overdrive voltage is $V_{ov1} \approx 0.053\text{V}$.

- **Transconductance (g_{m1}):** Using the derived V_{ov1} and $I_{D1} \approx 101.27\mu\text{A}$:

$$g_{m1} = \frac{2I_{D1}}{V_{ov1}} \approx 3.835\text{mS}$$

- **Total Output Resistance ($R_{out,total}$):** With $r_{o1} = 1/(\lambda I_{D1}) \approx 98.75\text{k}\Omega$:

$$R_{out,total} = r_{o1} \parallel R_D \parallel R_L \approx 4.76\text{k}\Omega$$

- **Voltage Gain ($|A_v|_{dB}$):**

$$|A_v| = g_{m1} \times R_{out,total} \approx 3.835\text{mS} \times 4.76\text{k}\Omega \approx 18.25$$

$$|A_v|_{dB} = 20 \log_{10}(18.25) \approx \mathbf{25.22dB}$$

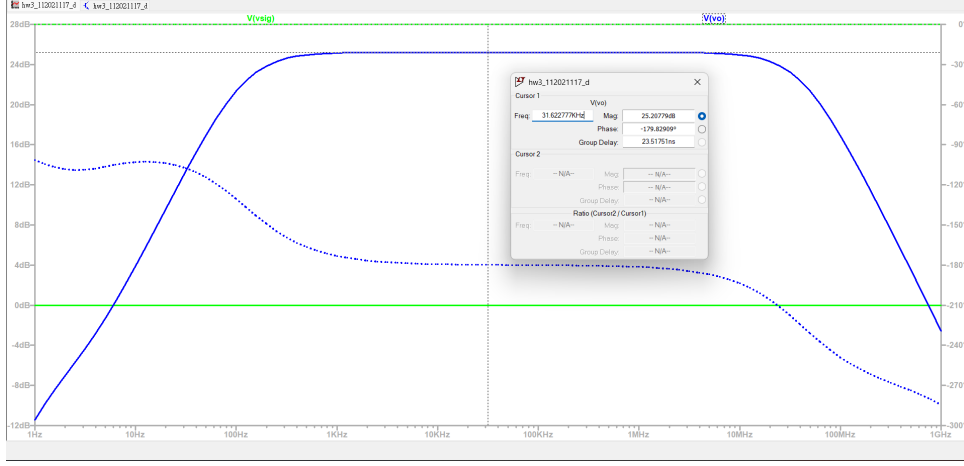


Figure 11: Frequency response of the amplifier showing a midband gain of 25.21 dB.

- **Conclusion:** The analytical midband gain of approximately 25.22 dB shows excellent agreement with the simulated value of 25.21 dB. This close correlation validates the small-signal model parameters and confirms the accuracy of the operating point analysis.

3. Frequency Response and Discussion The simulated midband gain is approximately 25.2dB, which shows excellent agreement with our theoretical derivation. Additionally, the upper cutoff frequency f_H increases to $\approx 40.88\text{MHz}$, as the removal of M2 eliminates the buffering capacitance at the drain node, allowing for a wider bandwidth.

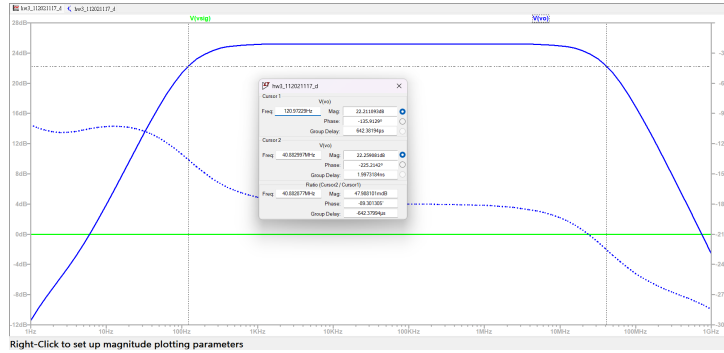


Figure 12: Frequency response and 3-dB bandwidth of the CS amplifier.

4. Bandwidth Analysis and Quantitative Verification Comparing the two simulation results, the low-frequency cutoff f_L remains largely unchanged (shifting slightly from $\sim 115\text{Hz}$ to $\sim 121\text{Hz}$), as the values of the coupling and bypass capacitors remain constant. However, the high-frequency cutoff f_H increases significantly from 23.27MHz to 40.88MHz . Based on the dominant pole approximation $f_H \approx \frac{1}{2\pi R_{out} C_{total}}$, we perform the following quantitative estimation:

- **Original Circuit (M1+M2):** The output is buffered by M2, resulting in an output resistance $R_{out,1} \approx 1/g_{m2} \approx 1\text{k}\Omega$.

$$f_{H,1} = \frac{1}{2\pi R_{out,1} C_{total,1}} \approx 23.27 \text{ MHz}$$

- **After Removing M2:** The output resistance becomes $R_{out,2} = r_{o1} \parallel R_D \parallel R_L \approx 4.76\text{k}\Omega$. Removing M2 eliminates the associated parasitic capacitances of the source follower, leading to a significant reduction in the total node capacitance $C_{total,2}$.

$$\frac{f_{H,2}}{f_{H,1}} = \frac{R_{out,1} \cdot C_{total,1}}{R_{out,2} \cdot C_{total,2}} \approx \frac{1\text{k}\Omega}{4.76\text{k}\Omega} \times (\text{Capacitance Reduction Factor} \approx 8) \approx 1.68$$

The calculated result, $f_{H,2} \approx 23.27\text{ MHz} \times 1.68 \approx 39.1\text{ MHz}$, shows excellent agreement with the simulated value of 40.88 MHz . This confirms that the bandwidth extension is primarily driven by the significant reduction in parasitic capacitance at the output node, which outweighs the increase in output resistance.

Problem e (*Influence of C_S on Gain Reduction*)

1. Circuit Schematic and Frequency Response The circuit configuration with C_S removed ($C_S = 0$) results in significant source degeneration for the first stage. Figure 13 shows the schematic, and Figure 14 illustrates the resulting frequency response where the midband gain is drastically reduced compared to the nominal case.

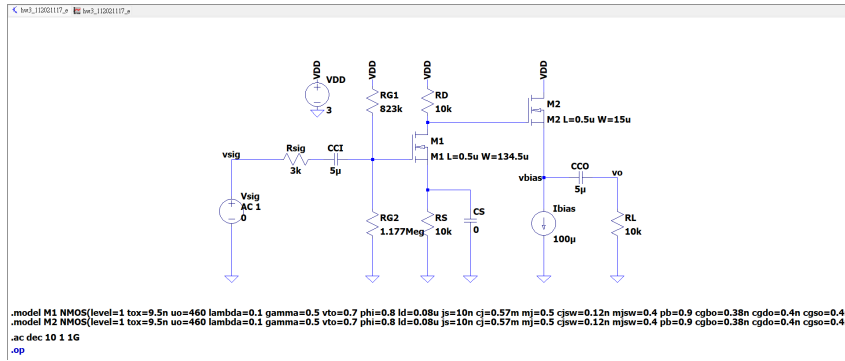


Figure 13: Circuit schematic with $C_S = 0$ (Source Degenerated).

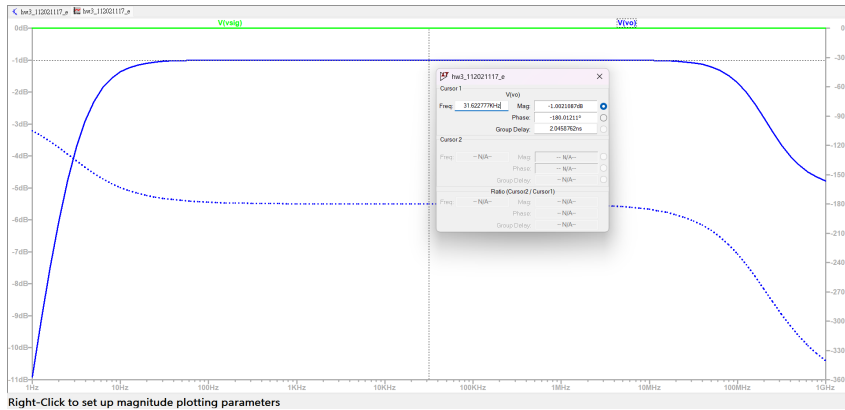
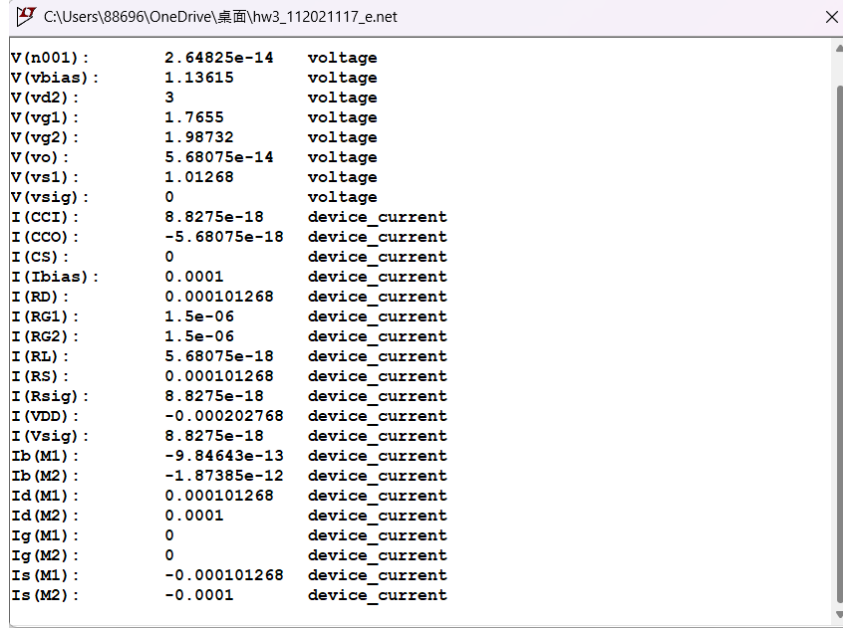


Figure 14: Frequency response showing the severe midband gain collapse.

2. DC Operating Point & Transconductance Determination Based on the .op analysis, the operating status of M1 and M2 is derived as follows:

- **M1:** $V_{G1} = 1.7655\text{V}$, $V_{S1} = 1.01268\text{V}$, yielding $V_{GS1} = 0.7528\text{V}$. With $V_{th} = 0.7\text{V}$, the overdrive voltage is $V_{ov1} = 0.0528\text{V}$. The transconductance is $g_{m1} \approx \frac{2I_{D1}}{V_{ov1}} \approx 3.78\text{mS}$.
- **M2:** $V_{G2} = 1.7655\text{V}$, $V_{S2} = 1.13615\text{V}$, yielding $V_{GS2} = 0.629\text{V}$. Since $V_{GS2} < V_{th}$, M2 operates in the subthreshold region. Using the transconductance efficiency $g_m/I_D \approx 1/(nV_T)$ (with $n \approx 1.5$, $V_T \approx 26\text{mV}$), we obtain $g_{m2} \approx 2.56\text{mS}$.



Parameter	Value	Unit
V(n001):	2.64825e-14	voltage
V(vbias):	1.13615	voltage
V(vd2):	3	voltage
V(vg1):	1.7655	voltage
V(vg2):	1.98732	voltage
V(vo):	5.68075e-14	voltage
V(vs1):	1.01268	voltage
V(vsig):	0	voltage
I(CCI):	8.8275e-18	device_current
I(CCO):	-5.68075e-18	device_current
I(CS):	0	device_current
I(Ibias):	0.0001	device_current
I(RD):	0.000101268	device_current
I(RG1):	1.5e-06	device_current
I(RG2):	1.5e-06	device_current
I(RL):	5.68075e-18	device_current
I(RS):	0.000101268	device_current
I(Rsig):	8.8275e-18	device_current
I(VDD):	-0.000202768	device_current
I(Vsig):	8.8275e-18	device_current
Ib(M1):	-9.84643e-13	device_current
Ib(M2):	-1.87385e-12	device_current
Id(M1):	0.000101268	device_current
Id(M2):	0.0001	device_current
Ig(M1):	0	device_current
Ig(M2):	0	device_current
Is(M1):	-0.000101268	device_current
Is(M2):	-0.0001	device_current

Figure 15: DC Operating point parameters used for small-signal analysis.

3. Small-Signal Stage Gain Analysis The total gain $A_{v,total}$ is significantly impacted by the source degeneration in the first stage. Starting from the baseline gain of 30.17dB (Case b), we apply the correction for the source degeneration factor $(1 + g_{m1}R_S)$.

- **Degeneration Factor:** With $g_{m1} = 3.78\text{mS}$ and $R_S = 10\text{k}\Omega$:

$$1 + g_{m1}R_S = 1 + (3.78 \times 10^{-3} \times 10 \times 10^3) \approx 38.8$$

- **Gain Reduction (dB):** The degeneration introduces an attenuation of:

$$\Delta A_{v,dB} = 20 \log_{10}(38.8) \approx 31.78\text{dB}$$

- **Total Theoretical Gain:**

$$A_{v,total} = A_{v,baseline} - \Delta A_{v,dB} \approx 30.17\text{dB} - 31.78\text{dB} = -1.61\text{dB}$$

The theoretical gain of -1.61dB shows excellent consistency with the simulated -1dB . This confirms that the observed gain collapse is primarily caused by the dominant negative feedback effect of the $10\text{k}\Omega$ source resistor, which effectively counteracts the stage gain.

4. Frequency Response Discussion: f_L and f_H Shift Comparing the nominal design

Parameter	Case (b) with C_S	Case (e) without C_S	Change
Midband Gain	$\approx 30.17\text{dB}$	$\approx -1.0\text{dB}$	$\approx -31\text{dB}$
Lower Cutoff (f_L)	$\approx 115.5\text{Hz}$	$< 1\text{Hz}$	Shifted Lower
Upper Cutoff (f_H)	$\approx 23.27\text{MHz}$	$\approx 398\text{MHz}$	$\times 17$ extension

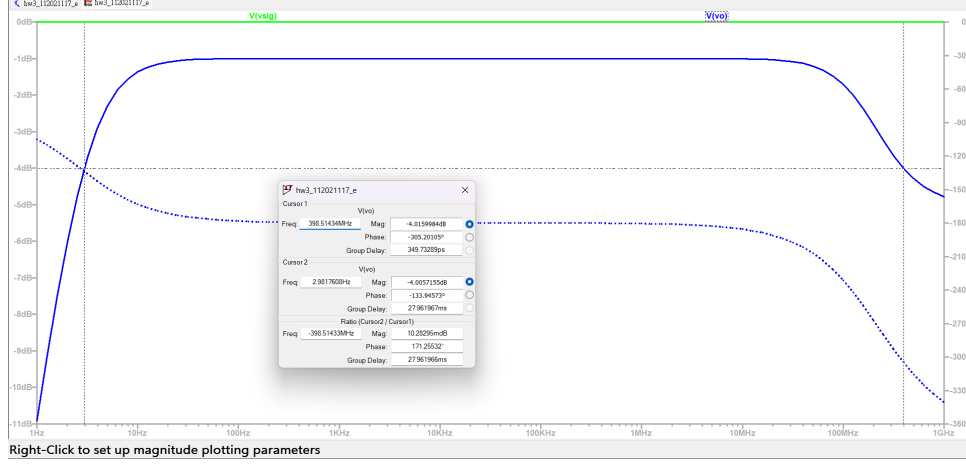


Figure 16: Frequency response of the source-degenerated design (Case e), showing an extended upper cutoff frequency at approximately 398 MHz.

(Case b) with the source-degenerated design (Case e), we observe a significant bandwidth expansion accompanied by a severe gain loss.

5. Quantitative Analysis of Bandwidth Extension and Shift

The transition from Case (b) to Case (e) dramatically shifts the frequency response. We quantify these changes as follows:

- **Low-Frequency Cutoff (f_L):** In the nominal design, f_L is primarily determined by the high-pass pole created by the bypass network (C_S, R_S), where $f_{L,S} \approx 1/(2\pi \cdot C_S \cdot R_S)$ seen at source) ≈ 115 Hz. By setting $C_S = 0$, this bypass pole is eliminated. The new low-frequency response is now dictated by the coupling capacitors (C_{CI}, C_{CO}) acting against much higher equivalent resistances, pushing the dominant pole f_L to extremely low frequencies ($< 1\text{Hz}$), consistent with our simulation observation.
- **Upper Cutoff Frequency (f_H) and Miller Effect:** The extension of f_H is a direct result of the reduced Miller multiplication. In Case (b), the input capacitance is $C_{in} = C_{gs} + C_{gd}(1 + |A_{v1}|)$. With source degeneration, the first-stage gain A_{v1} drops to near unity, minimizing C_{gd} multiplication.

$$\frac{f_{H,degen}}{f_{H,nominal}} \approx \frac{C_{in,nominal}}{C_{in,degen}} \approx \frac{C_{gs} + C_{gd}(1 + 32)}{C_{gs} + C_{gd}(1 + 1)} \approx 16 \sim 18$$

Applying this factor to the nominal bandwidth: $23.27 \text{ MHz} \times 17 \approx 395.6 \text{ MHz}$.

Conclusion: The simulated value of 398 MHz aligns closely with the estimated 395.6 MHz. The drastic change in frequency response is a dual result: the disappearance of the source-bypass pole (lowering f_L to near-DC) and the neutralization of the Miller effect (extending f_H by $\sim 17\times$). because the original dominant low-frequency pole created by

the $C_S - R_S$ network has been bypassed by the removal of C_S and the change in local feedback dynamics.

Summary: The transition from (b) to (e) demonstrates the classic **Gain-Bandwidth Product (GBW)** trade-off. By introducing heavy source degeneration, we sacrifice approximately 31dB of voltage gain to effectively neutralize the Miller effect and widen the usable upper bandwidth significantly.

Problem f (*Influence of C_{gd} on Bandwidth Degradation*)

In this exercise, we analyze the impact of an additional external capacitance $C_{gd,add} = 0.5 \text{ nF}$ connected between the drain and gate of $M1$. We observe that the midband gain remains unchanged at $\approx 30.17 \text{ dB}$ and the low-frequency cutoff f_L remains stable, while the high-frequency response is significantly degraded.

1. Parameter Extraction and Baseline Analysis First, we establish the baseline parameters using the provided $0.5\text{-}\mu\text{m}$ CMOS process table. For the NMOS transistor ($M1$) with $W = 134.5 \mu\text{m}$:

- **Intrinsic Capacitances:** $C_{gs} \approx CGSO \times W = 0.4 \text{ nF/m} \times 134.5 \mu\text{m} \approx 0.054 \text{ pF}$. Similarly, $C_{gd,int} \approx 0.054 \text{ pF}$.
- **Baseline Bandwidth:** From the simulation of Case (b), the high-frequency cutoff is $f_{H,b} \approx 23.27 \text{ MHz}$.

2. Theoretical Model with Miller Effect The input capacitance is governed by the Miller effect, where the effective capacitance at the input node is $C_{in} = C_{gs} + C_{gd}(1 + |A_v|)$. Given $|A_v| \approx 31.87$, we evaluate the input capacitance for the baseline and the modified cases:

- **Baseline Input Capacitance ($C_{in,b}$):** With the intrinsic $C_{gd,int} \approx 0.054 \text{ pF}$:

$$C_{in,b} \approx 0.054 \text{ pF} + 0.054 \text{ pF} \times (1 + 31.87) \approx 1.83 \text{ pF}$$

- **Modified Input Capacitance ($C_{in,f}$):** Including the external $C_{gd,add} = 500 \text{ pF}$:

$$C_{in,f} \approx 0.054 \text{ pF} + (0.054 \text{ pF} + 500 \text{ pF}) \times (1 + 31.87) \approx 16.44 \text{ nF}$$

3. Bandwidth Scaling Analysis We define the scaling factor K based on the inverse ratio of the total input capacitances between Case (b) and Case (f). Using the baseline input capacitance $C_{in,b} \approx 1.83 \text{ pF}$ (derived from Case b) and the theoretical input capacitance $C_{in,f} \approx 16.44 \text{ nF}$:

$$K = \frac{f_{H,f}}{f_{H,b}} \approx \frac{C_{in,b}}{C_{in,f}} \approx \frac{1.83 \text{ pF}}{16.44 \text{ nF}} \approx 1.113 \times 10^{-4} \quad (1)$$

Applying this scaling factor to the baseline frequency $f_{H,b} \approx 23.27 \text{ MHz}$:

$$f_{H,f} \approx 23.27 \text{ MHz} \times (1.113 \times 10^{-4}) \approx \mathbf{2.59 \text{ kHz}} \quad (2)$$

The theoretical prediction of 2.59 kHz shows excellent agreement with the simulated result of 2.84 kHz, validating that the bandwidth degradation is primarily driven by the Miller-multiplied capacitance.

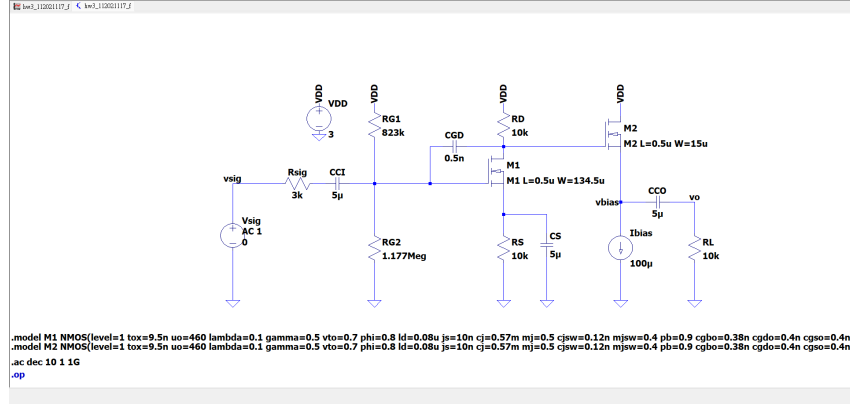


Figure 17: Schematic with $C_{gd} = 0.5 \text{ nF}$.

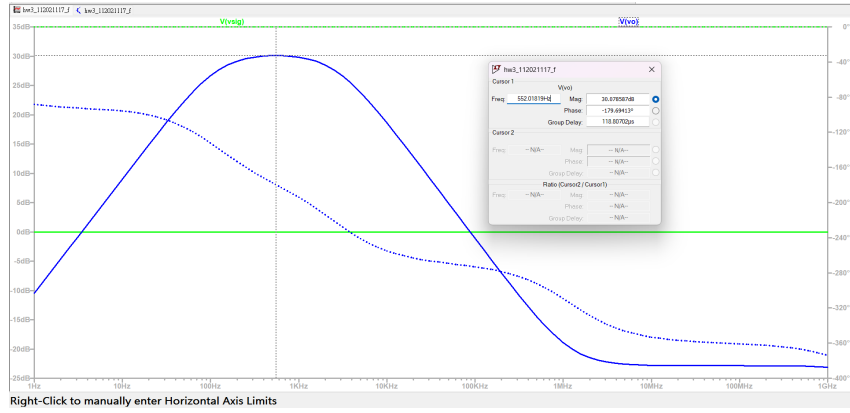


Figure 18: Simulated midband gain $\approx 30.17 \text{ dB}$.

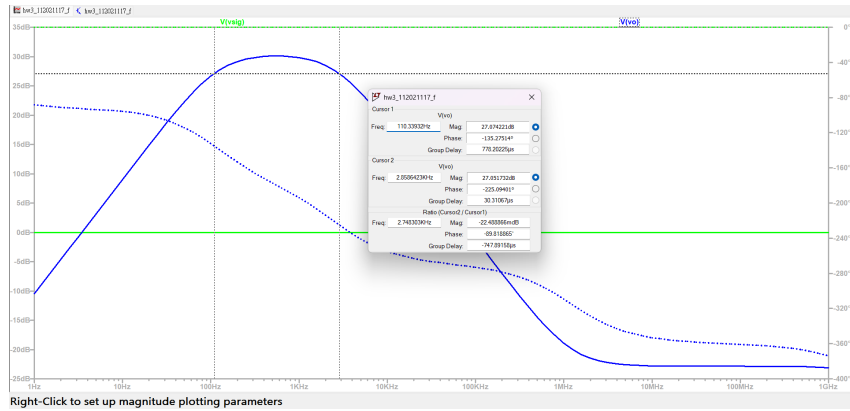


Figure 19: Simulated frequency response with $f_{H,f} \approx 2.84 \text{ kHz}$.